Hojin Kim

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Education

Purdue University — *IN, United States*
PhD in Mechanical Engineering — *Aug. 2025 – Present*
Advisor: Assistant Prof. Romit Maulik

The Pennsylvania State University — *PA, United States*
PhD in Informatics — *Aug. 2024 – Aug. 2025*
Advisor: Assistant Prof. Romit Maulik — *(Transfer to Purdue University)*

Seoul National University — *Seoul, Republic of Korea*
MS in Aerospace Engineering — *Mar. 2021 – Feb. 2023*
Advisor: Prof. Kwanjung Yee

Seoul National University — *Seoul, Republic of Korea*
BS in Aerospace Engineering — *Mar. 2015 – Feb. 2021*

Research Interest

Computational Fluid Dynamics (CFD), Machine learning, Turbulence, Optimization

Honors and Awards

Argonne Training Program on Extreme-Scale Computing (ATPESC) — *St. Charles, IL, USA*
Argonne National Laboratory, USA — *Jul. 2025 - Aug. 2025*
: Selected participant, intensive program on high-performance computing and scalable algorithms

Student Travel Award for the 2025 SIAM Conference on Computational Science and Engineering (CSE25) — *Jan. 2025*
Society for Industrial and Applied Mathematics (SIAM)

The Ruth Young Boucke Graduate Fellowship — *Aug. 2024*
The Robert W. Graham Endowed Graduate Fellowship
The Pennsylvania State University, PA, United States

Merit-based Scholarship — *Spring 2015, Spring 2022*
Seoul National University, Seoul, Republic of Korea

Work-study Scholarship — *Fall 2018, Spring 2019*
College of Engineering, Seoul National University, Seoul, Republic of Korea

Publication

H. Kim, R. Maulik (2025). "Towards Interpretable Deep Learning and Analysis of Dynamical Systems via the Discrete Empirical Interpolation Method," *under review, arXiv:2510.21852.*

H. Kim, V. Shankar, V. Viswanathan, R. Maulik (2024). "Generalizable data-driven turbulence closure modeling on unstructured grids with differentiable physics.," *under review, arXiv:2307.13533.*

S. Barwey, **H. Kim**, R. Maulik (2025). "Interpretable A-posteriori error indication for graph neural network surrogate models.," *Computer Methods in Applied Mechanics and Engineering*, 433, 117509.

H. Kim, J. Lee, D. Lee, K. Yee (2025). Improved Conceptual Design of eVTOL Aircraft: Considering Rotor–Rotor Interactional Effects. *International Journal of Aeronautical and Space Sciences*, 1-20.

S. Yang, **H. Kim**, Y. Hong, K. Yee, R. Maulik, N. Kang (2024). "Data-driven physics-informed neural networks: A digital twin perspective.," *Computer Methods in Applied Mechanics and Engineering*, 428, 117075.

Conference

18th U.S. National Congress On Computational Mechanics *Chicago, IL, USA*
S. Barwey, **H. Kim**, R. Maulik *Jul. 2025*
"Scalable, interpretable, and explainable scientific machine learning with geometric deep learning"

2025 SIAM Conference on Computational Science and Engineering *Fort Worth, TX, USA*
H. Kim, R. Maulik *Mar. 2025*
"Differentiable physics for generalizable closure modeling of separated flows"

APS 77th Annual Division of Fluid Dynamics Meeting *Salt Lake City, UT, USA*
H. Kim, R. Maulik *Nov. 2024*
"Differentiable physics for generalizable closure modeling of separated flows"

48th European Rotorcraft Forum *Winterthur, Switzerland*
H. Kim, J. Seo, D. Lee, Y. Hong, K. Yee *Sep. 2022*
"Design Methodology of Urban Air Mobility for Noise Mitigation at the Conceptual Design Stage"

Workshops

2025 Purdue CCAM workshop - Scientific Machine Learning: Theory, Algorithms, and Applications *West Lafayette, IN, USA* *Sep. 2025*
H. Kim, R. Maulik
"Differentiable physics for generalizable closure modeling of separated flows"

2025 AAAI Bridge Program - Knowledge-Guided Machine Learning *Philadelphia, PA, USA*
H. Kim, S. Barwey, R. Maulik *Feb. 2025*
"Scalable, adaptive, and explainable scientific machine learning with applications to surrogate models of partial differential equations"

2024 AAAI Symposium on Integrated Approaches to Computational Scientific Discovery *Arlington, VA, USA* *Nov. 2024*
H. Kim, S. Barwey, R. Maulik
"Scalable, adaptive, and explainable scientific machine learning with applications to surrogate models of partial differential equations"

Technologies

Languages: Python, C++, Fortran
Tools: HPC, Linux